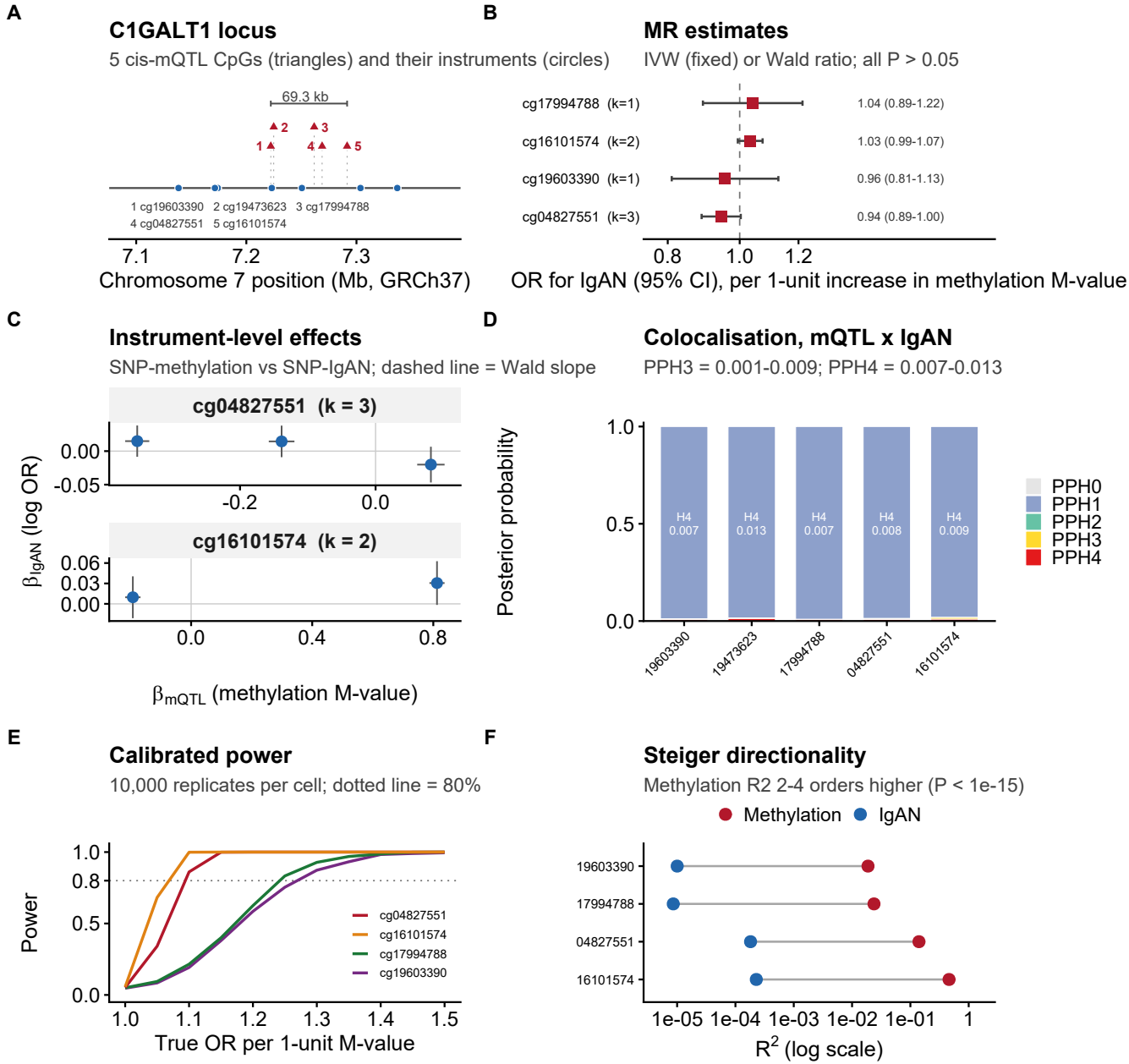

A
C1GALT1 locus
5 cis-mQTL CpGs (triangles) and their instruments (circles)
69.3 kb
1 cg19603390 2 cg19473623 3 cg17994788
4 cg04827551 5 cg16101574
7.1
7.2
7.3
Chromosome 7 position (Mb, GRCh37)
B
MR estimates
IVW (fixed) or Wald ratio; all P > 0.05
cg17994788 (k=1)
cg16101574 (k=2)
cg19603390 (k=1)
cg04827551 (k=3)
1.04 (0.89-1.22)
1.03 (0.99-1.07)
0.96 (0.81-1.13)
0.94 (0.89-1.00)
0.8
1.0
1.2
OR for IgAN (95% CI), per 1-unit increase in methylation M-value
C
Instrument-level effects
SNP-methylation vs SNP-IgAN; dashed line = Wald slope
cg04827551 (k = 3)
cg16101574 (k = 2)
βIgAN (log OR)
βmQTL (methylation M-value)
D
Colocalisation, mQTL x IgAN
PPH3 = 0.001-0.009; PPH4 = 0.007-0.013
Posterior probability
H4 0.007
H4 0.013
H4 0.007
H4 0.008
H4 0.009
19603390
19473623
17994788
04827551
16101574
PPH0
PPH1
PPH2
PPH3
PPH4
E
Calibrated power
10,000 replicates per cell; dotted line = 80%
Power
True OR per 1-unit M-value
cg04827551
cg16101574
cg17994788
cg19603390
F
Steiger directionality
Methylation R2 2-4 orders higher (P < 1e-15)
Methylation
IgAN
19603390
17994788
04827551
16101574
R2 (log scale)